

Erdős Problem #889

EP-889

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

For $k \geq 0$ and $n \geq 1$ let $v(n, k)$ count the prime factors of $n + k$ which do not divide $n + i$ for $0 \leq i < k$. Equivalently, $v(n, k)$ counts the number of prime factors of $n + k$ which are $> k$.

Is it true that $v_0(n) = \max_{k \geq 0} v(n, k) \rightarrow \infty$ as $n \rightarrow \infty$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>